

Void and sphere plasmons

Here, we derive the equations in Radiative decay of plasmons in a metallic nanoshell

([https://link.aps.org/pdf/10.1103/PhysRevB.69.155402?](https://link.aps.org/pdf/10.1103/PhysRevB.69.155402?casa_token=ysrofqx3MhMAAAAA:PO2xtgoR8T6w5glk_0FlmUHMIk5X6qtD1tdgkH6TqaDN3p0EXLMs_s-OFhfZkDREXNWQ3-83KUphDw)

[casa_token=ysrofqx3MhMAAAAA:PO2xtgoR8T6w5glk_0FlmUHMIk5X6qtD1tdgkH6TqaDN3p0EXLMs_s-OFhfZkDREXNWQ3-83KUphDw](https://link.aps.org/pdf/10.1103/PhysRevB.69.155402?casa_token=ysrofqx3MhMAAAAA:PO2xtgoR8T6w5glk_0FlmUHMIk5X6qtD1tdgkH6TqaDN3p0EXLMs_s-OFhfZkDREXNWQ3-83KUphDw))

In the region outside the sphere, we take

$\mathbf{H} = h_l^1(k_0 r) \mathbf{L}Y_{lm}(\theta, \phi)$, with $k_0 = \omega/c$ (purely outgoing wave—i.e. a quasinormal mode). Within the conducting shell, we have

$\mathbf{H} = (ah_l^1(kr) + bh_l^2(kr)) \mathbf{L}Y_{lm}(\theta, \phi)$, with $k = nk_0$

We must have $h_l^1(\rho_a) = ah_l^1(\eta_a) + bh_l^2(\eta_a)$ (where $\eta_a, \eta_b, \rho_a, \rho_b$ are defined in the paper; we refer to this condition as **Condition 1**) as well as

$\varepsilon(\omega)(\rho_a h_l^1(\rho_a))' = a(\eta_a h_l^1(\eta_a))' + b(\eta_a h_l^2(\eta_a))'$ (**Condition 2**)

We also have $cj_l(\rho_b) = ah_l^1(\eta_b) + bh_l^2(\eta_b)$ (**Condition 3**) and

$c\varepsilon(\omega)(\rho_b j_l(\rho_b))' = a(\eta_b h_l^1(\eta_b))' + b(\eta_b h_l^2(\eta_b))'$ (**Condition 4**)

Therefore, by multiplying **Condition 3** by $(\eta_b h_l^2(\eta_b))'$ and subtracting **Condition 4** multiplied by $h_l^2(\eta_b)$, we obtain (**Condition 5**):

$$a[(\eta_b h_l^2(\eta_b))' h_l^1(\eta_b) - (\eta_b h_l^1(\eta_b))' h_l^2(\eta_b)] = c[j_l(\rho_b)(\eta_b h_l^2(\eta_b))' - \varepsilon(\omega) h_l^2(\eta_b)(\rho_b j_l(\rho_b))']$$

Similarly, by multiplying **Condition 3** by $(\eta_b h_l^1(\eta_b))'$ and subtracting from it **Condition 4** multiplied by $h_l^1(\eta_b)$, we obtain (**Condition 6**):

$$b[h_l^2(\eta_b)(\eta_b h_l^1(\eta_b))' - h_l^1(\eta_b)(\eta_b h_l^2(\eta_b))'] = c[j_l(\rho_b)(\eta_b h_l^1(\eta_b))' - \varepsilon(\omega)(\rho_b j_l(\rho_b))' h_l^1(\eta_b)]$$

Note that the coefficients of a and b in **Condition 5** and **Condition 6**, respectively, are the same but with a relative minus sign. We wish to find the ratio:

$$\varepsilon(\omega)(\rho_a h_l^1(\rho_a))' / h_l^1(\rho_a) = \varepsilon(\omega) \alpha_l^E(\omega) / \beta_l^E(\omega),$$

where the equality follows from the definitions of α_l^E and β_l^E in the paper. Note that this ratio is exactly the ratio of **Condition 2** to **Condition 1**, which is equal to (using **Condition 5** and **Condition 6**):

$$\frac{a(\eta_a h_l^1(\eta_a))' + b(\eta_a h_l^2(\eta_a))'}{a h_l^1(\eta_a) + b h_l^2(\eta_a)} = \frac{[j_l(\rho_b)(\eta_b h_l^2(\eta_b))' - \varepsilon(\omega) h_l^2(\eta_b)(\rho_b j_l(\rho_b))'](\eta_a h_l^1(\eta_a))' - [j_l(\rho_b)(\eta_b h_l^1(\eta_b))' - \varepsilon(\omega)(\rho_b j_l(\rho_b))' h_l^1(\eta_b)](\eta_a h_l^2(\eta_a))'}{[j_l(\rho_b)(\eta_b h_l^2(\eta_b))' - \varepsilon(\omega) h_l^2(\eta_b)(\rho_b j_l(\rho_b))'] h_l^1(\eta_a) - [j_l(\rho_b)(\eta_b h_l^1(\eta_b))' - \varepsilon(\omega)(\rho_b j_l(\rho_b))' h_l^1(\eta_b)] h_l^2(\eta_a)}$$

The numerator may be separated into terms proportional to the dielectric function and those that are not:

$$\varepsilon(\omega)(\rho_b j_l(\rho_b))' \left[h_l^1(\eta_b)(\eta_a h_l^2(\eta_a))' - h_l^2(\eta_b)(\eta_a h_l^1(\eta_a))' \right] - j_l(\rho_b) \left[(\eta_b h_l^1(\eta_b))'(\eta_a h_l^2(\eta_a))' - (\eta_b h_l^2(\eta_b))'(\eta_a h_l^1(\eta_a))' \right]$$

Similarly for the denominator,

$$\varepsilon(\omega)(\rho_b j_l(\rho_b))' \left[h_l^1(\eta_b) h_l^2(\eta_a) - h_l^2(\eta_b) h_l^1(\eta_a) \right] - j_l(\rho_b) \left[h_l^2(\eta_a)(\eta_b h_l^1(\eta_b))' - (\eta_b h_l^2(\eta_b))' h_l^1(\eta_a) \right]$$

To put the various terms in the form reported in the paper, we note that $h_l^2 = -h_l^l + 2j_l$. All the terms with products of two Hankel functions of the first kind vanish and we may write the numerator and denominator (up to an overall factor which cancels out in the ratio and is therefore unimportant):

$$\varepsilon(\omega)(\rho_b j_l(\rho_b))' \left[h_l^1(\eta_b)(\eta_a j_l(\eta_a))' - j_l(\eta_b)(\eta_a h_l^1(\eta_a))' \right] - j_l(\rho_b) \left[(\eta_b h_l^1(\eta_b))'(\eta_a j_l(\eta_a))' - (\eta_b j_l(\eta_b))'(\eta_a h_l^1(\eta_a))' \right] = -\varepsilon(\omega)(\rho_b j_l(\rho_b))' A_l + j_l(\rho_b) B_l = -\alpha_l^E,$$

for the numerator (where A_l and B_l are defined in the paper). And, for the denominator:

$$\varepsilon(\omega)(\rho_b j_l(\rho_b))' \left[h_l^1(\eta_b) j_l(\eta_a) - j_l(\eta_b) h_l^1(\eta_a) \right] - j_l(\rho_b) \left[j_l(\eta_a)(\eta_b h_l^1(\eta_b))' - (\eta_b j_l(\eta_b))' h_l^1(\eta_a) \right] = -\varepsilon(\omega)(\rho_b j_l(\rho_b))' C_l + j_l(\rho_b) D_l = -\beta_l^E / \varepsilon(\omega)$$

Therefore, the ratio of the numerator to the denominator is exactly $\varepsilon(\omega)\alpha_l^E/\beta_l^E$, as desired, giving us **Equation 2** of the paper.

Numerical methods for evaluation of higher angular momentum modes

For higher angular momenta, we run into issues in evaluating **Equation 2** of the paper if we are to just brute force use the spherical Bessel/Hankel functions. This happens, in particular, for $l = 3$. The origin of this is the large magnitudes of η_a and η_b . Since the Hankel functions have an exponential dependence on these quantities, $e^{i\eta_a}$, $e^{i\eta_b}$, this makes them numerically unstable to be used directly. The solution to this is to evaluate the products of the Hankel functions and their derivatives before evaluating them numerically. In particular, for the case of $l = 3$, we have:

$$h_3^1(z)e^{-iz} = (-i) \left[\frac{15}{z^3} - \frac{6}{z} \right] \frac{1}{z} - \left[\frac{15}{z^2} - 1 \right] \frac{1}{z}$$

$$h_3^2(z)e^{iz} = (i) \left[\frac{15}{z^3} - \frac{6}{z} \right] \frac{1}{z} - \left[\frac{15}{z^2} - 1 \right] \frac{1}{z}$$

Using these, we also get the derivatives:

$$(zh_3^1(z))'e^{-iz} = -i \left[-\frac{45}{z^4} + \frac{6}{z^2} \right] + \left[\frac{30}{z^3} \right] + \left[\frac{15}{z^3} - \frac{6}{z} \right] - i \left[\frac{15}{z^2} - 1 \right] = i \left[\frac{45}{z^4} - \frac{21}{z^2} + 1 \right] + \left[\frac{45}{z^3} - \frac{6}{z} \right]$$

$$(zh_3^2(z))'e^{iz} = i \left[-\frac{45}{z^4} + \frac{6}{z^2} \right] + \left[\frac{30}{z^3} \right] + \left[\frac{15}{z^3} - \frac{6}{z} \right] + i \left[\frac{15}{z^2} - 1 \right] =$$

$$i \left[-\frac{45}{z^4} + \frac{21}{z^2} - 1 \right] + \left[\frac{45}{z^3} - \frac{6}{z} \right]$$

Therefore, in any combination involving the products of Hankel functions, e.g.

$h_3^1(\eta_a)h_3^2(\eta_b)$, we write this as $(h_3^1(\eta_a)e^{-i\eta_a})(h_3^2(\eta_b)e^{i\eta_b})e^{i(\eta_a-\eta_b)}$. The first two terms are numerically stable, as they are just polynomials, and the last term is more numerically stable than $e^{i\eta_a}$ or $e^{i\eta_b}$ since, $|(\eta_a - \eta_b)/\eta_a| \approx H/R$, which is on the order of $1e^{-2}$ in the range of interest of the paper. Since this shows up in an exponent, a two order of magnitude difference has huge consequences.